

THE CENTRAL-CHARGE THRESHOLD FOR WALL-CROSSING IN TWO-VARIABLE OPEN FJRW THEORY

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ABSTRACT. Gross, Kelly and Tessler constructed genus-zero open FJRW invariants for $(x^r + y^s, \mu_r \times \mu_s)$ whose values can depend on boundary conditions. We prove that the primary invariants are independent of those choices exactly when the central charge is less than one, equivalently for the simple Fermat singularities A_{n-1} , D_4 , E_6 , and E_8 . The least number of primary interior insertions producing dependence satisfies an explicit trichotomy; two insertions first suffice for $x^4 + y^4$. By contrast, one interior insertion with one descendent produces dependence for every model.

If a critical graph, the combinatorial type indexing a codimension-one wall, has p and q boundary markings of the two types, its wall-crossing vector field is the Hamiltonian vector field of $x^p y^q$ for $dx \wedge dy$. The resulting determinant bracket identifies the wall-crossing Lie algebra with a graded subalgebra of the tropical vertex Lie algebra after an exponent shift. At central charge one the first critical graph has one boundary marking of each type, Hamiltonian xy , and vector field $x\partial_x - y\partial_y$. This one-dimensional torus direction is omitted from the wall-crossing algebra as stated by Gross–Kelly–Tessler. This is the first appearance of the simply-elliptic cases in the open theory. The threshold agrees with a more general result of Maher.

1. INTRODUCTION

Background. For a nondegenerate quasi-homogeneous polynomial W and a group G of diagonal symmetries, FJRW theory [FJR13] assigns to the Landau–Ginzburg model (W, G) a cohomological field theory: a genus-zero A -model built from moduli of W -spin curves and the Euler class of a Witten bundle. In their monograph [GKT22], Gross, Kelly and Tessler construct the *open* analogue for the two-variable Fermat model

$$W = x^r + y^s, \quad G = \mu_r \times \mu_s, \quad r, s \geq 2,$$

replacing closed W -spin curves by W -spin disks. The resulting open FJRW invariants are integrals of multisections of a descendent Witten bundle over moduli spaces that are real orbifolds with corners, and their central output is an open mirror theorem: a generating function of the open invariants is a versal deformation of W whose oscillatory period integrals recover the closed extended FJRW invariants, with the deformation parameters already flat [GKT22, Theorem 0.3].

A structural feature distinguishes the open theory from its closed counterpart and from the open r -spin theory of Buryak–Clader–Tessler [BCT24, BCT22]: the invariants are *not* canonical. Boundary conditions for the Witten bundle over the corners of moduli space are constrained, but not determined, by the positivity requirements of [GKT22]; the residual freedom makes the open invariants jump as the boundary data cross certain walls. Gross,

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Kelly and Tessler organize these jumps into a pro-nilpotent Landau–Ginzburg wall-crossing group $G^{r,s}$ acting on the set of possible systems of invariants, and prove that this action is faithful and transitive [GKT22, Theorems 0.7 and 4.27]: the systems of open invariants form a single torsor, while the period integrals – hence the closed theory – are constant along it.

The source of every wall is a *critical graph*: a stable W -spin graph Λ whose Witten bundle has rank one above the dimension of its moduli space, $\text{rank } E_\Lambda(\mathbf{d}) = \dim \overline{\mathcal{M}}_\Lambda^W + 1$ [GKT22, Definition 2.33]. Such a graph indexes a one-parameter family of codimension-one boundary strata through which zeros of the multisection can escape. The formally defined wall-crossing Lie algebra $\mathfrak{g}^{r,s}$ is a direct sum of one-dimensional pieces indexed by critical graphs of nonzero bidegree [GKT22, Definition 4.23]. The excluded bidegree $(0,0)$ is spanned by $x\partial_x - y\partial_y$. We adjoin that direction explicitly in (5), while keeping GKT’s original algebra and group distinct from the enlarged objects used here.

The prediction. Whether critical graphs exist depends on (r, s) and on whether one allows descendents. In the descendent-free (*primary*) theory, [GKT22, Remark 0.9] predicts that wall-crossing is absent exactly for the simple singularities, and asserts this for an explicit list of pairs. As stated, that remark contains a discrepancy between its list and its naming, which we record and correct below. The prediction itself – that the primary open FJRW invariants are canonical if and only if W is a simple singularity – is left open in [GKT22] and is reiterated as such in the survey [GKT26].

Results. We prove the prediction and recast it as a clean invariant-theoretic threshold. Recall that the *central charge* of $W = x^r + y^s$, whose variables carry weights $q_x = 1/r$ and $q_y = 1/s$, is

$$(1) \quad \hat{c}_W = (1 - 2q_x) + (1 - 2q_y) = 2 - \frac{2}{r} - \frac{2}{s}.$$

Fix a finite set I of labeled interior markings, each label i carrying a twist (a_i, b_i) with $0 \leq a_i \leq r - 2$ and $0 \leq b_i \leq s - 2$. A family of GKT canonical boundary conditions is called *symmetric* if it is invariant under permutations of labels with the same twist. For a nonempty subset $J \subseteq I$ and descendent degrees $\mathbf{d} = (d_i)_{i \in J}$, set

$$(2) \quad r(J) \equiv \sum_{i \in J} a_i \pmod{r}, \quad 0 \leq r(J) < r, \quad s(J) \equiv \sum_{i \in J} b_i \pmod{s}, \quad 0 \leq s(J) < s,$$

$$(3) \quad m(J, \mathbf{d}) = rs + \sum_{i \in J} (sa_i + rb_i + rs(d_i - 1)), \quad w(J, \mathbf{d}) = m(J, \mathbf{d}) - rs.$$

Here m is GKT’s mass and w is the *Euler weight*. Write $t_{a,b,d}$ for the deformation variable associated with an insertion $\tau_d^{(a,b)}$ and $u_{J,\mathbf{d}} = \prod_{i \in J} t_{a_i, b_i, d_i}$. The coefficient ring $A_{I, \text{sym}}$ is the symmetric GKT deformation ring in which the total multiplicity of each twist is bounded by the number of labels of that twist in I ; in particular every positive-degree element is nilpotent.

A critical graph with data $(J, \mathbf{d})$ and $k_1 + 1$, respectively $k_2 + 1$, boundary markings of the two types satisfies

$$(4) \quad sk_1 + rk_2 = w(J, \mathbf{d}).$$

GKT’s Lie algebra $\mathfrak{g}^{r,s}$ contains the corresponding generators when $(k_1, k_2) \neq (0, 0)$ but excludes bidegree $(0, 0)$ [GKT22, Def. 4.23]. We call a critical graph of Euler weight zero *marginal*; by (4) its vector field is $x\partial_x - y\partial_y$. The *marginally enlarged* algebra and group over

$A_{I,\text{sym}}$ are

$$(5) \quad \begin{aligned} \tilde{\mathfrak{g}}_{A_{I,\text{sym}}}^{r,s} &= \mathfrak{g}_{A_{I,\text{sym}}}^{r,s} \oplus \bigoplus_{\substack{\emptyset \neq J \subseteq I, \mathbf{d} \geq 0 \\ w(J, \mathbf{d})=0, r(J)=s(J)=0}} \mathbb{Q} u_{J, \mathbf{d}} (x \partial_x - y \partial_y), \\ \tilde{G}_{A_{I,\text{sym}}}^{r,s} &= \exp(\tilde{\mathfrak{g}}_{A_{I,\text{sym}}}^{r,s}). \end{aligned}$$

The exponential is finite over $A_{I,\text{sym}}$. This notation keeps GKT's original positive-weight group distinct from the enlargement used here.

Proposition 1.1 (Marginal enlargement of the GKT action). *The group (5) acts on the systems of symmetric canonical open FJRW invariants with marking set I , preserves that set, and acts transitively. The jump automorphisms in the proof of [GKT22, Thm. 5.9] belong to this enlarged group.*

Theorem 1.2 (Central-charge criterion). *Let $W = x^r + y^s$ with $r, s \geq 2$ and central charge $\hat{c}_W$ as in (1). The following are equivalent.*

- (i) $\hat{c}_W < 1$.
- (ii) $\frac{1}{r} + \frac{1}{s} > \frac{1}{2}$.
- (iii) W is a simple singularity: up to interchange of x and y ,
 $(r, s) \in \{(n, 2) : n \geq 2\} \cup \{(3, 3), (3, 4), (3, 5)\},$

of type A_{n-1}, D_4, E_6, E_8 respectively.

- (iv) *For every finite labeled marking set I as above, the genus-zero primary open FJRW invariants of $(x^r + y^s, \mu_r \times \mu_s)$ are independent of the choice of symmetric canonical boundary conditions.*

When these fail, that is when $\hat{c}_W \geq 1$, there is a finite marking set I for which the primary invariants genuinely depend on the boundary conditions.

The equivalence (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) is elementary and classical; the content is (iv). The threshold $\hat{c}_W = 1$ is exactly the boundary between the simple singularities ($\hat{c}_W < 1$) and the parabolic (simple-elliptic, $\hat{c}_W = 1$) and hyperbolic ($\hat{c}_W > 1$) regimes. Thus the presence of primary wall-crossing detects, through open enumerative geometry, the classical ADE threshold.

Beyond the dichotomy, the same count determines *when* wall-crossing first appears. For an interior insertion the relevant twists lie in the range $0 \leq a \leq r - 2, 0 \leq b \leq s - 2$; write n for the number of interior insertions.

Theorem 1.3 (First wall). *Let $n_*(r, s)$ be the least number of primary interior insertions admitting a critical graph. Then*

$$n_*(r, s) = \begin{cases} \infty, & \frac{1}{r} + \frac{1}{s} > \frac{1}{2} \quad (\text{simple}), \\ 2, & \min(r, s) \geq 4, \\ 3, & \min(r, s) = 3 \text{ and } \max(r, s) \geq 6. \end{cases}$$

The case $n_* = 2$ is first realized by $W = x^4 + y^4$ with two interior insertions of twist $(2, 2)$, the example of [GKT26, §6].

Finally, descendents change the picture entirely: they generate walls in every case, including the simple singularities, as [GKT22, Remark 0.9] anticipates. The first such wall is universal.

Proposition 1.4 (First descendent wall). *For every (r, s) with $r, s \geq 2$, the data of a single interior insertion of descendent degree one admit exactly one critical graph. Hence the open FJRW invariants with descendents exhibit wall-crossing for every two-variable Fermat model, simple or not.*

Where walls do occur, their generators have a uniform Hamiltonian description. Gross, Kelly and Tessler write the wall-crossing Lie algebra in terms of the vector fields $X_{a,b} = x^a y^b ((b+1)x\partial_x - (a+1)y\partial_y)$ [GKT22, eqn. (4.19)] and observe that they preserve $dx \wedge dy$ [GKT22, p. 89]. We name the Hamiltonian and read off the consequence. The bracket below is computed for the affine-plane form $dx \wedge dy$; after the exponent shift by $(1, 1)$ it is the usual determinant bracket for the torus form $d \log x \wedge d \log y$ (Remark 5.4).

Proposition 1.5 (Hamiltonian wall generators). *For $\omega = dx \wedge dy$ and $H_{a,b} = x^{a+1}y^{b+1}$, the wall generator $X_{a,b}$ is the Hamiltonian vector field of $H_{a,b}$; in particular it is divergence-free. The assignment $H_{a,b} \mapsto X_{a,b}$ is an isomorphism of graded Lie algebras from the Poisson algebra on the monomials $x^p y^q$, $p, q \geq 1$, onto its image, under which the bracket is the tropical-vertex determinant*

$$[X_{a,b}, X_{c,d}] = ((b+1)(c+1) - (a+1)(d+1)) X_{a+c, b+d}.$$

It restricts to an isomorphism onto the wall-crossing Lie algebra $\mathfrak{g}^{r,s}$ of [GKT22] after retaining the positive-weight critical Hamiltonians, and onto $\tilde{\mathfrak{g}}^{r,s}$ after also adjoining the marginal Hamiltonian xy . Thus the enlarged algebra is the graded Hamiltonian subalgebra spanned by the critical Hamiltonians $\{H_\Lambda = x^{k_1+1}y^{k_2+1}\}$, where the critical graph Λ has k_1+1 and k_2+1 boundary marked points of the two types, so that H_Λ is obtained by recording those two boundary-marking counts as the exponents of x and y . The (s, r) -weighted degree of H_Λ equals $w(\Lambda) + r + s$, with w the Euler weight of (3); marginal graphs ($H_\Lambda = xy$) contribute the adjoined torus direction of (5). This makes the tropical-vertex analogy of [GKT22, Remarks 0.8 and 4.20] an explicit map.

Proposition 1.5 completes the program the criterion was part of: an ADE/central-charge classification of when walls exist, a determination of the first wall, and a Hamiltonian normal form for the wall generators. The one structure it does *not* supply — a slope-ordered factorization whose rays are indexed by the shifted boundary degrees $(k_1 + 1, k_2 + 1)$ of critical graphs — is constructed in the companion paper [Wue26], for which the classification here is the starting point; see §5 and the closing section.

Method. Nothing here rebuilds the machinery of [GKT22]. Theorem 1.2 combines the closed-form count of critical graphs [GKT22, Notation 2.40], GKT’s positive-weight wall algebra and faithful action [GKT22, Definition 4.23 and Theorem 4.27], and the local marginal extension of Proposition 1.1. The arithmetic core is a single inequality, Lemma 3.1, bounding the graph count by $n \hat{c}_W$. All numerical claims (the ADE list, the trichotomy $n_*(r, s)$, the first-wall classification with its marginal/positive-weight split, and the universal descendent wall) are proved in §§3–6; finite exact checks are described in §7.

Relation to prior work. The uniqueness of the primitive form for simple singularities is classical [Sai83, NY98], and is the closed-mirror shadow of the statement that these are the potentials without wall-crossing. On the open side, Basalaev–Buryak constructed open Saito potentials for the A - and D -series by solving the open WDVV equations [BB21]; the present criterion explains, uniformly, why the primary open theory is unambiguous on the whole ADE locus, including the E -cases, for which no closed-form potential has yet been computed. The wall-crossing group of [GKT22] is drawn, “in spirit,” by analogy with the

tropical vertex group of Kontsevich–Soibelman [KS06] and its role in Fano mirror symmetry [Gro10] ([GKT22, Remark 0.8]); separately, a tropical version of open FJRW theory is raised as an open problem in [GKT26, §6], there through tropical moduli. The classification of critical graphs organized here is a prerequisite for a scattering-diagram form combining the two, carried out in [Wue26]. After this note was drafted we learned that Maher’s thesis establishes the threshold in substantially greater generality, for invertible polynomials, via the (non)triviality of the primary wall-crossing group [Mah24, Proposition 1.0.5]; we keep the present treatment for its elementary Fermat-case proof, the explicit first-wall classification, and the Hamiltonian normal form, on which the scattering construction builds directly.

2. THE COUNT OF CRITICAL GRAPHS

We recall only what is needed and refer to [GKT22] for all definitions. Fix $W = x^r + y^s$. A genus-zero smooth (r, s) -disk carries interior marked points with twists (a_i, b_i) , $0 \leq a_i \leq r - 1$, $0 \leq b_i \leq s - 1$, and a descendent vector $\mathbf{d} = (d_i) \in \mathbb{Z}_{\geq 0}$. The interior insertions contributing to the primary potential lie in the Neveu–Schwarz range $0 \leq a_i \leq r - 2$, $0 \leq b_i \leq s - 2$ [GKT22, eqn. (0.5)].

Given a set J of interior markings with twists $(a_i, b_i)_{i \in J}$ and descendents $\mathbf{d} = (d_i)_{i \in J}$, define, following [GKT22, Notation 2.30],

$$(6) \quad \sum_{i \in J} a_i = r(J) + \ell_1 r, \quad \sum_{i \in J} b_i = s(J) + \ell_2 s,$$

where $r(J) \in \{0, \dots, r - 1\}$ and $s(J) \in \{0, \dots, s - 1\}$ are the residues, so that

$$(7) \quad \ell_1 = \left\lfloor \frac{1}{r} \sum_{i \in J} a_i \right\rfloor, \quad \ell_2 = \left\lfloor \frac{1}{s} \sum_{i \in J} b_i \right\rfloor.$$

The classification of balanced and critical graphs with interior markings exactly J is [GKT22, Notation 2.40]: setting

$$(8) \quad N(J, \mathbf{d}) = \ell_1 + \ell_2 - |J| + 1 + \sum_{i \in J} d_i,$$

one has $|\text{Balanced}(J, \mathbf{d})| = N(J, \mathbf{d}) + 1$ and

$$(9) \quad |\text{Crit}(J, \mathbf{d})| = \max(N(J, \mathbf{d}), 0).$$

Here the balanced graphs are the disk types whose invariants occur as coefficients of the open potential; the critical graphs index the possible codimension-one jumps between those coefficients. In particular a critical graph with interior markings J and descendent vector $\mathbf{d}$ exists if and only if $N(J, \mathbf{d}) \geq 1$. (Throughout, $J \neq \emptyset$. For $J = \emptyset$, the balanced equation has precisely the solutions $(k_1, k_2) = (r, 0), (0, s)$, the two boundary-only graphs appearing in [GKT22, Example 3.4]; the critical equation forces $(0, 0)$, whose two boundary markings are unstable. Thus (9) is not asserted in the boundary-only case.)

Proof of Proposition 1.1. In the proof of [GKT22, Thm. 5.9], the set $Q_0(A, \mathbf{d})$ indexes precisely the interior marking data whose critical strata acquire isolated zeros during a homotopy of boundary conditions. Its dimension constraint is (4); hence the resulting jump elements consist of GKT’s positive-weight summands together with the marginal summands in (5).

It remains to check preservation and transitivity on realizable systems. The chamber-index equations of [GKT22, Def. 3.47] are Gamma-weighted polynomial relations among the balanced-disk coefficients; they encode the period and open-recursion constraints, together with

the one-marking normalizations. Take $\hbar < 0$ and expand, coefficientwise, the positive-quadrant Laplace integral

$$\mathcal{L}_\hbar(W) = \int_0^\infty \int_0^\infty e^{W/\hbar} dx dy.$$

Its monomial moments are Gamma factors; consequently its $u_{J,\mathbf{d}}$ -coefficient is a fixed nonzero multiple of GKT's polynomial $\mathcal{A}(J, \mathbf{d}, \nu)$. Every enlarged generator is divergence-free and tangent to the coordinate axes (Lemma 5.1), so integration by parts gives $\mathcal{L}_\hbar(e^\nu W) = \mathcal{L}_\hbar(W)$. Thus all the $\mathcal{A}$ -conditions are preserved, including the residue classes not detected by the distinguished oscillatory periods. The normalizations in Definition 3.47 are also preserved: a primary critical monomial has at least two markings, and a descendent singleton does not change the primary normalization. The realization theorem [GKT22, Thm. 5.13 and Cor. 5.14] identifies the solutions of these equations with the systems arising from canonical boundary conditions, which proves preservation.

For transitivity, let S be a finite set of deformation monomials that contains every divisor of each of its elements, and pass to the quotient that retains exactly the coefficients indexed by S . Order S by a linear extension of proper divisibility. Fix $u = u_{J,\mathbf{d}} \in S$ and write $N = N(J, \mathbf{d})$. The balanced disk types over $(J, \mathbf{d})$ are indexed by $0 \leq p \leq N$ and have boundary degrees

$$(r(J) + pr, s(J) + (N - p)s).$$

Let $V_u = \bigoplus_{p=0}^N \mathbb{Q}e_p$ be their coefficient space, where e_p changes only the coefficient of the p th balanced type. Once two systems agree on every proper divisor of u , every term of $\mathcal{A}(J, \mathbf{d}, -)$ involving a partition into two or more deformation monomials already agrees. Their difference in V_u therefore lies in the kernel of the remaining one-part term, a nonzero linear functional $L_u : V_u \rightarrow \mathbb{Q}$ whose coefficients are the Gamma factors computed above. Thus $\dim \ker L_u = N$.

For $1 \leq p \leq N$, the critical graph between the $(p-1)$ st and p th balanced types acts on V_u by

$$w_p = a_p e_p - b_p e_{p-1}, \quad a_p = r(s(J) + (N - p)s + 1), \quad b_p = s(r(J) + (p - 1)r + 1).$$

Both coefficients are positive. The Laplace-invariance calculation above gives $L_u(w_p) = 0$, while reading successively the $e_N, e_{N-1}, \dots, e_1$ coordinates shows that the w_p are independent. They therefore form a basis of $\ker L_u$. If the diagonal is marginal, the weight equation forces $N = 1$ and $r(J) = s(J) = 0$; its only wall vector is $w_1 = re_1 - se_0$, precisely the torus vector adjoined in (5). Hence in every case a unique combination of the critical-graph generators with coefficient u removes the difference in V_u . Acting by that combination can alter only coefficients indexed by strict multiples of u , so it does not disturb earlier steps. Induction proves transitivity in the quotient defined by S . These corrections are compatible when S is enlarged, and the symmetric formal coefficient ring is the inverse limit of these finite divisor-closed quotients; passing to that limit proves transitivity over $A_{I,\text{sym}}$. $\square$

The reduction from invariants to graph counts is the following consequence of [GKT22], which we isolate for use in both directions of Theorem 1.2.

Proposition 2.1. *Fix a finite marking set $I \subseteq \mathbb{N}$ and work over the coefficient ring $A_{I,\text{sym}} = \mathbb{Q}[t_{a,b,d}]/\text{Ideal}(I)$ of [GKT22] (the ideal caps each twist (a,b) at the number of labels of I carrying it). Let $\mathbf{d} = \mathbf{0}$.*

- (a) *If no critical graph with descendent vector $\mathbf{0}$ has interior markings contained in I , then $\widehat{G}_{A_{I,\text{sym}}}^{r,s}$ fixes every primary open FJRW invariant; the primary invariants are canonical.*

- (b) *If some critical graph with descendent vector $\mathbf{0}$ has interior markings contained in I , then two systems of symmetric canonical open FJRW invariants over I differ in a primary invariant.*

Proof. By [GKT22, Definition 4.23], the original wall-crossing Lie algebra $\mathfrak{g}_{A_{I,\text{sym}}}^{r,s}$ is spanned by the summands corresponding to critical graphs with $(k_1, k_2) \neq (0, 0)$. Equation (5) adds the omitted $(0, 0)$ summands. After passage to symmetric coefficients, critical graphs with the same twist–descendent multiset determine the same summand. For a critical graph with interior markings J , descendents $\mathbf{d}$, and boundary degrees (k_1, k_2) , its generator is

$$(10) \quad v_\Lambda = u_{J,\mathbf{d}} x^{k_1} y^{k_2} ((k_2 + 1)x\partial_x - (k_1 + 1)y\partial_y), \quad u_{J,\mathbf{d}} = \prod_{i \in J} t_{a_i, b_i, d_i}.$$

Here (k_1, k_2) is the bidegree used in [GKT22, Definition 4.23]: the critical graph has $k_1 + 1$ boundary marked points of type σ_1 and $k_2 + 1$ of type σ_2 [GKT22, eqn. (2.31), Proposition 2.39], and the index satisfies the critical condition $sk_1 + rk_2 = w(J, \mathbf{d})$ of (4). Positive weight gives a summand of $\mathfrak{g}^{r,s}$, while $(k_1, k_2) = (0, 0)$ gives the adjoined marginal torus direction. GKT’s group $G_{A_{I,\text{sym}}}^{r,s} = \exp(\mathfrak{g}_{A_{I,\text{sym}}}^{r,s})$ acts on the chamber indices $\Upsilon_{r,s}^{\text{sym}}(I)$ of [GKT22, Definition 3.47] by $g(W^\nu) = W^{g(\nu)}$ [GKT22, Theorem 4.27]; Proposition 1.1 extends the action transitively to $\tilde{G}_{A_{I,\text{sym}}}^{r,s}$.

For (a): if there is no $\mathbf{0}$ -critical graph, every generator (10) has some $d_i > 0$, hence $u_{J,\mathbf{d}}$ is divisible by a variable $t_{a,b,d}$ with $d > 0$. Acting on a potential $W^\nu = x^r + y^s + \sum_{0 \leq a \leq r-2, 0 \leq b \leq s-2} t_{a,b,0} x^a y^b + \cdots$ [GKT22, eqn. (0.5)], such a generator raises the total degree in the positive-descendent variables and therefore leaves unchanged the coefficients of the purely primary monomials $u_{J',\mathbf{0}} x^\alpha y^\beta$. Since each primary open invariant $\nu_{\Gamma,\mathbf{0}}$, the coefficient attached to a graph Γ in [GKT22, eqns. (0.4)–(0.5)], is, up to the fixed nonzero sign and automorphism factor in the symmetric potential, such a coefficient, it is fixed by every generator, hence by $\tilde{G}_{A_{I,\text{sym}}}^{r,s}$. Transitivity of the enlarged action then makes these primary invariants equal in every system.

For (b): a $\mathbf{0}$ -critical graph Λ furnishes a nonzero generator v_Λ with $u_{J,\mathbf{0}} = \prod_{i \in J} t_{a_i, b_i, 0}$ a monomial in primary variables only. Fix any system ν of symmetric canonical invariants, which exists by [GKT22, Theorem 0.1]. The coefficient of the deformation monomial $u_{J,\mathbf{0}}$ in $\exp(v_\Lambda)(W^\nu) - W^\nu$ is exactly $v_\Lambda(x^r + y^s)$, namely

$$u_{J,\mathbf{0}} (r(k_2 + 1)x^{k_1+r}y^{k_2} - s(k_1 + 1)x^{k_1}y^{k_2+s}),$$

which is nonzero. Proposition 1.1 ensures that $\exp(v_\Lambda) \cdot \nu$ is another system of symmetric canonical invariants over I . The two systems therefore differ in a primary invariant. $\square$

3. PROOF OF THE CENTRAL-CHARGE CRITERION

The equivalences (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) are immediate: (1) gives $\hat{c}_W < 1 \Leftrightarrow \frac{2}{r} + \frac{2}{s} > 1$, and $\frac{1}{r} + \frac{1}{s} > \frac{1}{2}$ characterizes the two-variable Fermat simple singularities, whose only types are $A_{n-1} = (n, 2)$, $D_4 = (3, 3)$, $E_6 = (3, 4)$ and $E_8 = (3, 5)$ (there is no Fermat E_7 , as $E_7 = x^3 + xy^3$ is not of the form $x^r + y^s$; likewise $D_n = x^{n-1} + xy^2$ is Fermat only for $n = 4$, where $x^3 + y^3 \cong D_4$). It remains to prove (i) $\Leftrightarrow$ (iv), for which, by Proposition 2.1, it suffices to decide the existence of a primary ($\mathbf{d} = \mathbf{0}$) critical graph. The whole matter rests on one inequality.

Lemma 3.1. *Let J be a set of $n = |J|$ interior markings with Neveu–Schwarz twists, $0 \leq a_i \leq r - 2$ and $0 \leq b_i \leq s - 2$. Then, with ℓ_1, ℓ_2 as in (7),*

$$\ell_1 + \ell_2 \leq n \hat{c}_W.$$

Consequently the primary count (8) obeys $N(J, \mathbf{0}) \leq 1 - n(1 - \hat{c}_W)$.

Proof. Dropping the floor functions and using the twist bounds,

$$\ell_1 + \ell_2 = \left\lfloor \frac{1}{r} \sum a_i \right\rfloor + \left\lfloor \frac{1}{s} \sum b_i \right\rfloor \leq \frac{\sum a_i}{r} + \frac{\sum b_i}{s} \leq \frac{n(r-2)}{r} + \frac{n(s-2)}{s} = n \left(2 - \frac{2}{r} - \frac{2}{s} \right) = n \hat{c}_W.$$

The second claim is $N(J, \mathbf{0}) = \ell_1 + \ell_2 - n + 1 \leq n \hat{c}_W - n + 1$. $\square$

Proof of (i) $\Rightarrow$ (iv). Assume $\hat{c}_W < 1$. For any J with $n = |J| \geq 1$ interior markings, Lemma 3.1 gives $\ell_1 + \ell_2 \leq n \hat{c}_W < n$; as $\ell_1 + \ell_2$ is an integer strictly below the integer n , we get $\ell_1 + \ell_2 \leq n - 1$, i.e. $N(J, \mathbf{0}) \leq 0$. By (9) there is no primary critical graph, for any J and any finite I . Proposition 2.1(a) then says the primary invariants are canonical for every I . (The case $n = 0$ is vacuous: there is no interior marking to twist, and $N = 1$ corresponds to the two boundary-only balanced graphs of [GKT22, Example 3.4], which are not critical.) $\square$

Proof of $\neg(i) \Rightarrow \neg(iv)$. Assume $\hat{c}_W \geq 1$, i.e. $\frac{2}{r} + \frac{2}{s} \leq 1$; in particular $r, s \geq 3$ and not both equal to 3, so $r, s \geq 3$ with $\max(r, s) \geq 4$, and we may take all interior twists maximal, $(a_i, b_i) = (r - 2, s - 2)$. With n such markings,

$$(11) \quad N = \left\lfloor \frac{n(r-2)}{r} \right\rfloor + \left\lfloor \frac{n(s-2)}{s} \right\rfloor - n + 1 = \left(n - \left\lceil \frac{2n}{r} \right\rceil \right) + \left(n - \left\lceil \frac{2n}{s} \right\rceil \right) - n + 1 = n - \left\lceil \frac{2n}{r} \right\rceil - \left\lceil \frac{2n}{s} \right\rceil + 1,$$

using $\lfloor n(r-2)/r \rfloor = \lfloor n - 2n/r \rfloor = n - \lceil 2n/r \rceil$. We exhibit n with $N \geq 1$, i.e. $\lceil 2n/r \rceil + \lceil 2n/s \rceil \leq n$.

If $\frac{2}{r} + \frac{2}{s} < 1$ (the hyperbolic regime $\hat{c}_W > 1$), then $\lceil 2n/r \rceil + \lceil 2n/s \rceil \leq \frac{2n}{r} + \frac{2n}{s} + 2 = n(\frac{2}{r} + \frac{2}{s}) + 2$, which is $\leq n$ once $n(1 - \frac{2}{r} - \frac{2}{s}) \geq 2$; such n exists since $1 - \frac{2}{r} - \frac{2}{s} > 0$.

If $\frac{2}{r} + \frac{2}{s} = 1$ (the parabolic regime $\hat{c}_W = 1$: the pairs $(4, 4)$, $(3, 6)$, $(6, 3)$), choose n with $r \mid 2n$ and $s \mid 2n$, e.g. $n = \text{lcm}(r, s)/2$, an integer for all three parabolic pairs; then $\lceil 2n/r \rceil + \lceil 2n/s \rceil = \frac{2n}{r} + \frac{2n}{s} = n$, so $N = 1$.

In either case a primary critical graph exists once I contains n interior labels of maximal twist, which is possible because each twist is realized infinitely often in the marking data [GKT22, §2.4]. By Proposition 2.1(b) the primary invariants over such an I depend on the boundary conditions. This proves $\neg(iv)$, and with it Theorem 1.2. $\square$

Remark 3.2 (On Remark 0.9 of [GKT22]). [GKT22, Remark 0.9] states that the pairs $(r, s) \in \{(n, 2), (2, n), (3, 3), (3, 4), (4, 3)\}$ give simple singularities “ A_{n-1} , E_6 , and E_8 ”. Two adjustments reconcile the list with the naming. First, $(3, 3)$ appears in the list but D_4 is not named; $x^3 + y^3$ is the D_4 singularity (three distinct lines, Milnor number 4). Second, E_8 is named but its pairs $(3, 5)$, $(5, 3)$ are absent from the list; these are $x^3 + y^5$, Milnor number 8. The corrected wall-crossing-free set is

$$\{(n, 2), (2, n) : n \geq 2\} \cup \{(3, 3), (3, 4), (4, 3), (3, 5), (5, 3)\},$$

of types A_{n-1}, D_4, E_6, E_8 , exactly as in Theorem 1.2(iii). The theorem shows this list is not merely a set of examples but the complete answer to the prediction.

4. THE FIRST WALL AND THE DESCENDENT WALL

Proof of Theorem 1.3. The simple case $n_* = \infty$ is Theorem 1.2. Assume $\hat{c}_W \geq 1$ and use (11), which gives the maximal N over size- n primary multisets, since N is nondecreasing in the twists and maximal at $(r - 2, s - 2)$. Thus $n_*(r, s)$ is the least n with $\lceil 2n/r \rceil + \lceil 2n/s \rceil \leq n$.

$\min(r, s) \geq 4$. For $n = 1$, $\lceil 2/r \rceil + \lceil 2/s \rceil = 1 + 1 = 2 > 1$ (each $2/r, 2/s \leq \frac{1}{2}$). For $n = 2$ and $r, s \geq 4$, $\lceil 4/r \rceil + \lceil 4/s \rceil = 1 + 1 = 2 = n$, so $N = 1$ and $n_* = 2$.

$\min(r, s) = 3$, say $r = 3$, with $\max = s \geq 6$. Here $a_i \leq 1$, so $\ell_1 = \lfloor n/3 \rfloor$. For $n = 2$, $\lceil 4/3 \rceil + \lceil 4/s \rceil = 2 + 1 = 3 > 2$. For $n = 3$, $\lceil 6/3 \rceil + \lceil 6/s \rceil = 2 + 1 = 3 = n$ (using $s \geq 6$), so $N = 1$ and $n_* = 3$. For $3 \leq s \leq 5$ the pair is simple and $n_* = \infty$, already covered.

Finally $\min(r, s) = 2$ forces the A -type simple case, $n_* = \infty$. This exhausts all (r, s) . For $x^4 + y^4$, the witnessing multiset is two insertions of twist $(r - 2, s - 2) = (2, 2)$, giving $N = 1$: the example of [GKT26, §6]. $\square$

Proof of Proposition 1.4. Take $|J| = 1$ with any twist. Then $\ell_1 = \ell_2 = 0$ and, by (8), $N(J, \mathbf{d}) = 0 + 0 - 1 + 1 + d = d$, where d is the descendent degree of the single insertion. Hence $d = 1$ gives $N = 1$: exactly one critical graph, for every (r, s) . Proposition 2.1(b) is stated for $\mathbf{d} = \mathbf{0}$, but its proof uses only preservation by the enlarged action and the exact coefficient computation. Applied to the descendent-1 generator, that computation produces a monomial divisible by $t_{a,b,1}$ with nonzero coefficient, so two systems of invariants differ in a descendent invariant. Hence the descendent theory wall-crosses in every case. $\square$

Remark 4.1. The trichotomy of Theorem 1.3 shows the $x^4 + y^4$ example of [GKT26] is not isolated: two interior insertions already force a wall for every (r, s) with $r, s \geq 4$, and three suffice whenever $\min(r, s) = 3 \leq \frac{1}{2} \max(r, s)$. Combined with Proposition 1.4, the qualitative picture is complete: the descendent-free theory is canonical exactly on the ADE locus, while switching on a single descendent breaks canonicity everywhere.

5. THE WALL GENERATORS ARE HAMILTONIAN

We now prove Proposition 1.5. Throughout, $\omega = dx \wedge dy$, and for a bivariate polynomial H the Hamiltonian vector field is $X_H := (\partial_y H) \partial_x - (\partial_x H) \partial_y$, so that $\iota_{X_H} \omega = dH$. In the rank-two case the marginally enlarged wall-crossing Lie algebra $\widetilde{\mathfrak{g}}^{r,s}$ is a sum of one-dimensional spaces spanned by

$$(12) \quad X_{a,b} = x^a y^b ((b+1)x \partial_x - (a+1)y \partial_y) = (b+1)x^{a+1} y^b \partial_x - (a+1)x^a y^{b+1} \partial_y,$$

one for each critical graph, the exponents $(a, b) = (k_1, k_2)$ being the bidegree used in [GKT22, Definition 4.23]: the graph has k_1+1 and k_2+1 boundary marked points of the two types, so the Hamiltonian $H_{k_1, k_2} = x^{k_1+1} y^{k_2+1}$ of Lemma 5.1 below is its boundary monomial.

Lemma 5.1. *For all $a, b \geq 0$, $X_{a,b} = X_{H_{a,b}}$ with $H_{a,b} = x^{a+1} y^{b+1}$. In particular $\operatorname{div} X_{a,b} = 0$, so $X_{a,b}$ preserves ω .*

Proof. $\partial_y H_{a,b} = (b+1)x^{a+1} y^b$ and $\partial_x H_{a,b} = (a+1)x^a y^{b+1}$, so $X_{H_{a,b}} = (b+1)x^{a+1} y^b \partial_x - (a+1)x^a y^{b+1} \partial_y$, which is (12). A Hamiltonian vector field is divergence-free: $\operatorname{div} X_{a,b} = \partial_x \partial_y H_{a,b} - \partial_y \partial_x H_{a,b} = 0$, recovering the volume-preservation of [GKT22, p. 89]. $\square$

Lemma 5.2. *For all $a, b, c, d \geq 0$,*

$$(13) \quad [X_{a,b}, X_{c,d}] = ((b+1)(c+1) - (a+1)(d+1)) X_{a+c, b+d}.$$

Equivalently, with the Poisson bracket $\{H, K\} := X_H(K)$ induced by ω ,

$$\{H_{a,b}, H_{c,d}\} = ((b+1)(c+1) - (a+1)(d+1)) H_{a+c, b+d}.$$

Proof. Write $X_{a,b} = f \partial_x + g \partial_y$ with $f = (b+1)x^{a+1} y^b$, $g = -(a+1)x^a y^{b+1}$, and $X_{c,d} = F \partial_x + G \partial_y$ similarly. A direct computation of $[X_{a,b}, X_{c,d}] = (X_{a,b}F - X_{c,d}f) \partial_x + (X_{a,b}G - X_{c,d}g) \partial_y$ gives, for the ∂_x -coefficient,

$$X_{a,b}F - X_{c,d}f = (b+d+1)[(b+1)(c+1) - (a+1)(d+1)] x^{a+c+1} y^{b+d},$$

which is $((b+1)(c+1) - (a+1)(d+1))$ times the ∂_x -coefficient $(b+d+1)x^{a+c+1}y^{b+d}$ of $X_{a+c,b+d}$; the ∂_y -coefficient matches identically. For the Poisson form, $\{H_{a,b}, H_{c,d}\} = X_{a,b}(H_{c,d}) = (b+1)(c+1)x^{a+c+1}y^{b+d+1} - (a+1)(d+1)x^{a+c+1}y^{b+d+1}$, and $X_{H_{a+c,b+d}} = X_{a+c,b+d}$ by Lemma 5.1 shows the two identities agree. $\square$

The determinant $\langle(a,b), (c,d)\rangle = (b+1)(c+1) - (a+1)(d+1)$ appearing in (13) is the structure constant of the tropical vertex Lie algebra of Kontsevich–Soibelman [KS06]; see [Gro10]. Assembling Lemmas 5.1 and 5.2 proves Proposition 1.5: $H_{a,b} \mapsto X_{a,b}$ is a Lie-algebra isomorphism from the Poisson algebra on $\{x^p y^q : p, q \geq 1\}$ onto its image, restricting to an isomorphism onto $\tilde{\mathfrak{g}}^{r,s}$ once one keeps only the Hamiltonians H_Λ of critical graphs. Restricting to positive weight recovers GKT’s $\mathfrak{g}^{r,s}$. The grading is immediate from the critical condition.

Proposition 5.3. *Let Λ be a critical graph with interior markings J , descendents $\mathbf{d}$, and index (k_1, k_2) , and set $m(\Lambda) = m(J, \mathbf{d}) = rs + \sum_{i \in J} (sa_i + rb_i + rs(d_i - 1))$ [GKT22, Notation 2.30]. Then the Hamiltonian $H_\Lambda = x^{k_1+1}y^{k_2+1}$ has (s, r) -weighted degree*

$$s(k_1 + 1) + r(k_2 + 1) = m(\Lambda) + r + s - rs.$$

Thus $\tilde{\mathfrak{g}}^{r,s}$ is graded by the (s, r) -weighted degree of the wall Hamiltonian, and the degree records the mass of the critical graph (equivalently its Euler weight $w(\Lambda) = m(\Lambda) - rs$ of (3)).

Proof. The critical condition is $sk_1 + rk_2 = m(\Lambda) - rs$ [GKT22, Proposition 2.39]; add $r + s$. $\square$

Remark 5.4 (Relation to the tropical vertex algebra). The tropical vertex Lie algebra is the Hamiltonian algebra of the *torus* form $d \log x \wedge d \log y$, for which $\{x^u, x^v\} = \langle u, v \rangle x^{u+v}$ with $\langle u, v \rangle$ the determinant, so that Hamiltonians add exponents. Proposition 1.5 uses instead the *affine-plane* form $dx \wedge dy = xy (d \log x \wedge d \log y)$, under which $\{x^u, x^v\} = \langle u, v \rangle x^{u+v-(1,1)}$. The two Poisson structures, hence the two Lie algebras, are intertwined by the exponent shift $u \mapsto u - (1, 1)$ induced by this factor xy – the same change of primitive form that makes $dx \wedge dy$ the natural volume in [GKT22, Theorem 0.3]. So $\tilde{\mathfrak{g}}^{r,s}$ is an affine-plane realization of the tropical-vertex bracket, restricted to critical Hamiltonians. This makes explicit the analogy that [GKT22, Remarks 0.8 and 4.20] draw “in spirit” with the Kontsevich–Soibelman group. The companion paper [Wue26] groups the corresponding automorphisms by the rays spanned by $(k_1 + 1, k_2 + 1)$, orders those ray factors by slope, and uses their products to transport the open potentials. No result of the present note depends on that construction. The tropical open FJRW theory called for in [GKT26, §6] is posed there through tropical moduli, a priori a distinct route to the same tropical picture.

6. THE FIRST WALL AT THE PARABOLIC BOUNDARY

The threshold $\hat{c}_W = 1$ is not only where wall-crossing begins across the family; the walls it produces have a uniform shape. By the *first wall* of (r, s) we mean the set of critical graphs over interior multisets of the minimal size $n_*(r, s)$. Combining the count (9) with the Hamiltonian dictionary of §5 gives a complete classification of the first wall for each (r, s) ; at the parabolic boundary it collapses to a single wall.

Proposition 6.1 (Parabolic first wall). *For any interior multiset J with $N(J, \mathbf{0}) = 1$, the index of its unique critical graph is $(k_1, k_2) = (r(J), s(J))$, of Euler weight $sr(J) + rs(J)$. For the three parabolic pairs, $(r, s) \in \{(4, 4), (3, 6), (6, 3)\}$ with $\hat{c}_W = 1$, the first wall consists of exactly one critical graph, its multiset forced to be the maximal-twist one: two copies of $(2, 2)$ for $(4, 4)$, three copies of $(1, 4)$ for $(3, 6)$, three copies of $(4, 1)$ for $(6, 3)$. Its residues*

vanish, so its index is $(0,0)$, its weight is zero, and its direction is the marginal torus direction $x\partial_x - y\partial_y$ of (5), with Hamiltonian xy . By contrast, for the hyperbolic pairs of Table 1 the first wall comprises several critical graphs, both genuine positive-weight walls and marginal torus directions.

Proof. If $N(J, \mathbf{0}) = 1$ there is a unique critical graph and, by [GKT22, Notation 2.40, eqn. (2.31)] read in the indexing of (10), its index is $(k_1, k_2) = (r(J), s(J))$; the weight statement is the critical condition (4). For each parabolic pair the minimal critical multiset is uniquely the maximal-twist one: at $n = n_*$ the inequality $\ell_1 + \ell_2 \geq n$ is attained only with equality and only at $(a_i, b_i) = (r-2, s-2)$ (for $(4,4)$, $\ell_1, \ell_2 \leq 1$ force $a_i = b_i = 2$; for $(3,6)$, $\ell_1 \leq 1, \ell_2 \leq 2$ force $a_i = 1, b_i = 4$). There the residues vanish, so $(k_1, k_2) = (0,0)$: the torus direction, with Hamiltonian $H_{0,0} = xy$ (Lemma 5.1). The hyperbolic data is the direct enumeration recorded in Table 1. $\square$

(r, s)	$\hat{c}_W$	regime	n_*	graphs	marginal	wall Hamiltonians
$(4, 4)$	1	parabolic	2	1	1	— (torus xy only)
$(3, 6)$	1	parabolic	3	1	1	— (torus xy only)
$(6, 3)$	1	parabolic	3	1	1	— (torus xy only)
$(4, 5)$	11/10	hyperbolic	2	2	1	xy^2
$(3, 7)$	22/21	hyperbolic	3	2	1	xy^2
$(4, 6)$	7/6	hyperbolic	2	4	2	xy^2, xy^3
$(5, 5)$	6/5	hyperbolic	2	5	2	xy^2, x^2y, x^2y^2

TABLE 1. The first wall of $x^r + y^s$: the number of critical graphs at $n = n_*(r, s)$, how many are marginal (index $(0,0)$, the torus direction with Hamiltonian xy), and the boundary monomials $x^{k_1+1}y^{k_2+1}$ Hamiltonian to the genuine wall generators. At the parabolic boundary the first wall is purely marginal; genuine walls appear only in the hyperbolic regime. Computed exactly; see §7.

The parabolic pairs are the two-variable Fermat *simply-elliptic* singularities $(x^4 + y^4 = \tilde{E}_7, x^3 + y^6 = \tilde{E}_8)$, for which the primitive form ceases to be $dx \wedge dy$ and becomes a period of an elliptic curve; the marginal torus direction is where that transition first registers in the open theory: the elliptic wall-crossing group of [Mah24, §5.2.2] is exactly this torus factor, and the explicit jumping of the $x^4 + y^4$ invariants in [GKT26, Example 4.9] is its action. Genuine (pro-nilpotent) walls appear precisely in the hyperbolic regime. The first *symmetric* hyperbolic Fermat pair is $x^5 + y^5$, with boundary monomials xy^2, x^2y, x^2y^2 in Table 1; its wall functions and scattering diagram are computed in [Wue26].

7. FINITE EXACT CHECKS

The proofs above are uniform in r and s ; the following finite calculations serve only as independent reproducibility checks. Substitution in (8) for every $2 \leq r, s \leq 40$ reproduces the ADE list and the trichotomy of Theorem 1.3. The absence assertion outside this finite range is supplied by Lemma 3.1, not by enumeration. The same calculation reproduces the first-wall data printed in Table 1, including the zero-weight/positive-weight distinction, and the single-descendent indices of Proposition 1.4. Finally, direct polynomial differentiation

verifies Lemma 5.1 and identity (13) for symbolic a, b, c, d ; evaluation for $0 \leq a, b, c, d \leq 4$ provides a finite numerical check.¹

CONCLUDING REMARKS AND OPEN DIRECTIONS

The companion paper [Wue26] continues the Hamiltonian identification by grouping the automorphisms on rays of boundary degree and factoring them in slope order. Two further directions remain open for the threshold problem itself.

- (1) *Explicit ADE potentials.* On the wall-crossing-free locus the primary potential is unambiguous; computing it in closed form for D_4, E_6, E_8 (matching the D -series potentials of [BB21] and extending them to the E -cases) and relating the coefficients to Coxeter/root-system data is a concrete computational program.
- (2) *The parabolic boundary.* Proposition 6.1 pins the first wall of the parabolic pairs to the marginal torus direction. What remains is the analytic side: these are the simply-elliptic singularities $\tilde{E}_7, \tilde{E}_8$, whose primitive form is a period of an elliptic curve, so the flat structure is governed by modular forms rather than polynomials [Mah24, §4.2.2]. Identifying the modular period map on the torus factor, beyond the $x^4 + y^4$ jumping of [GKT26, Example 4.9], is the first genuinely transcendental case of the theory.

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¹The exact-arithmetic files are `src/census.py` and `src/gkt_algebra.py` in the repository cited in [Wue26].